Project Group Information

Our project is called Wardrobe. The team members are Nick Nogueira, Reid Maury, and Alexis Chastang.

We will be using the following GitHub repository for our project:
https://github.com/Nicolas-Nogueira/WARDROBE

Project Description

Wardrobe is a web application that allows users to pick clothing from a catalog to generate outfits. The app helps users build a digital closet and easily create outfits from the pieces they already own.

Wardrobe is designed for anyone interested in fashion, whether they are deeply into style, just starting to explore fashion, or simply want to dress better and feel more confident in their outfits. It is especially useful for people who want to be more intentional with what they wear and make better use of their existing wardrobe. We are looking to make a website that is mobile-friendly as well as easy and straightforward to use.

Background and Motivation

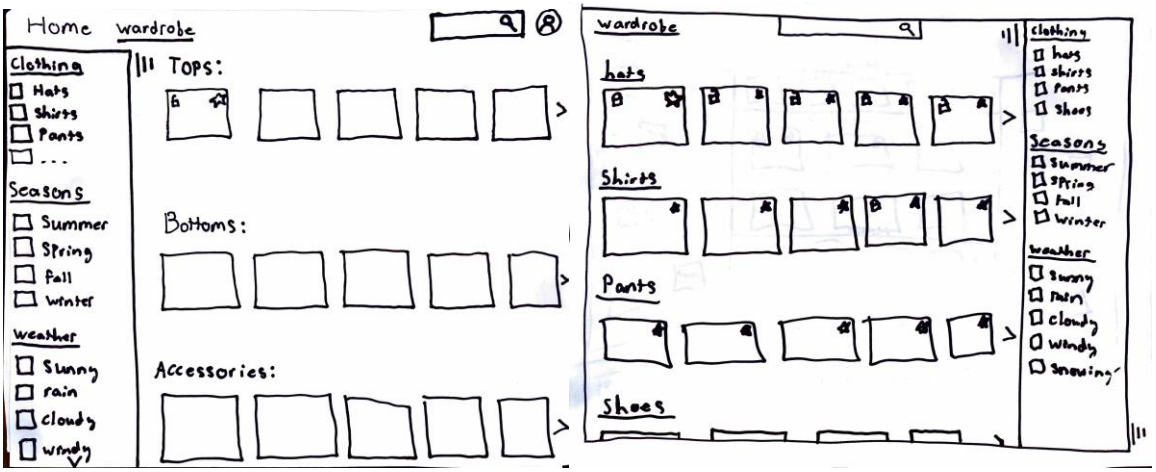
We were drawn to this project because it solves frustration, not knowing what to wear. Many people struggle with choosing an outfit, especially when they are short on time. It is easy to fall into the habit of wearing the same clothes repeatedly or feeling overwhelmed by too many options. Wardrobe helps solve this problem by taking the guesswork and effort out of picking an outfit, making the process faster and more intentional.

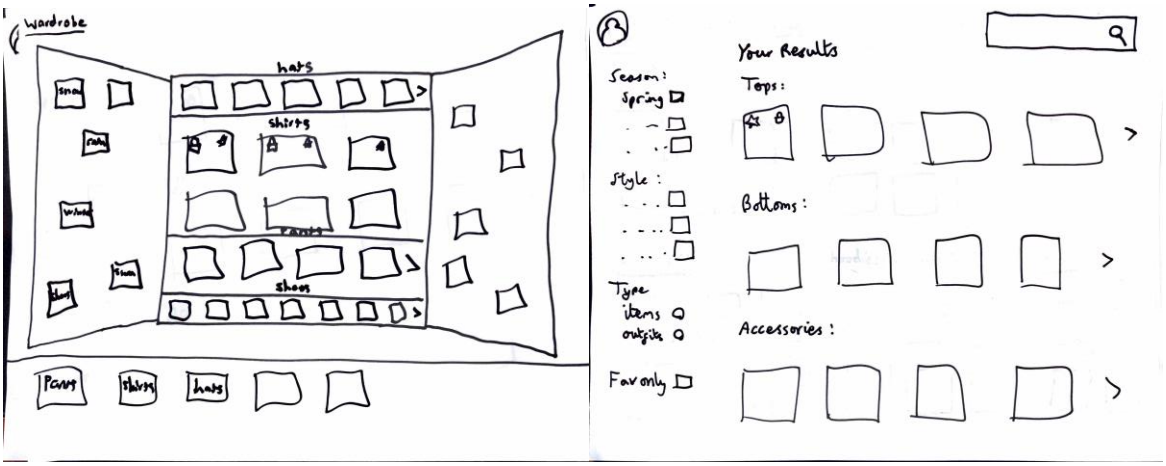
We are excited to work on a full-stack application over the course of the semester. We are looking forward to learning more about backend development and the database aspect of the project. Building a system that stores, categorizes, and retrieves user clothing data will help us strengthen our skills in web development and software design.

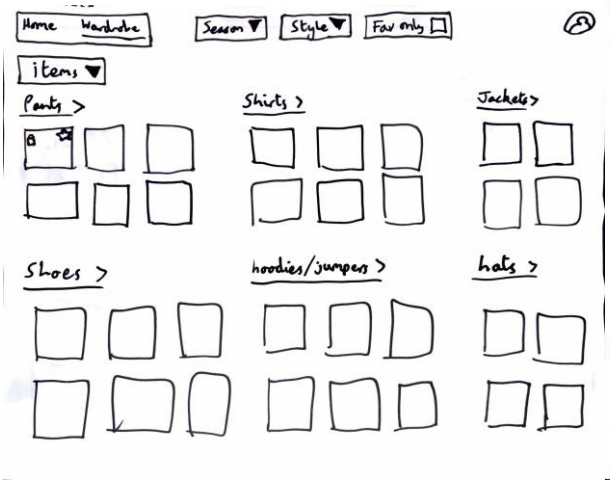
Overall, we are passionate about creating something practical, useful, and technically challenging, which will help us carry through the semester.

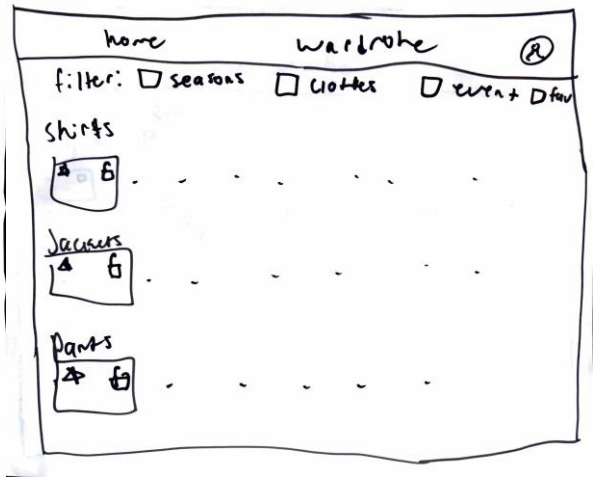
Wireframes and Prototyping

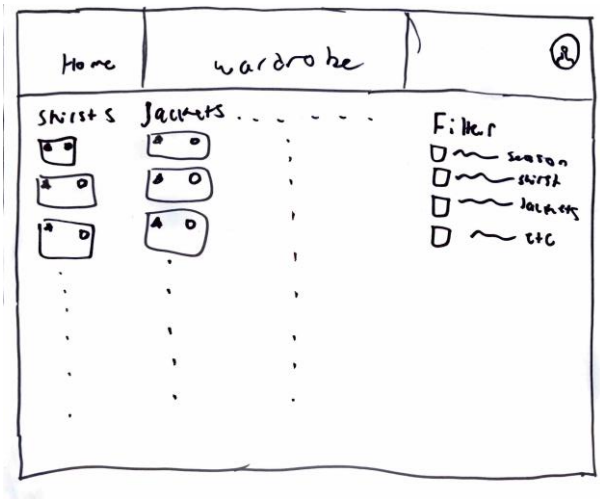
Wardrobe page:



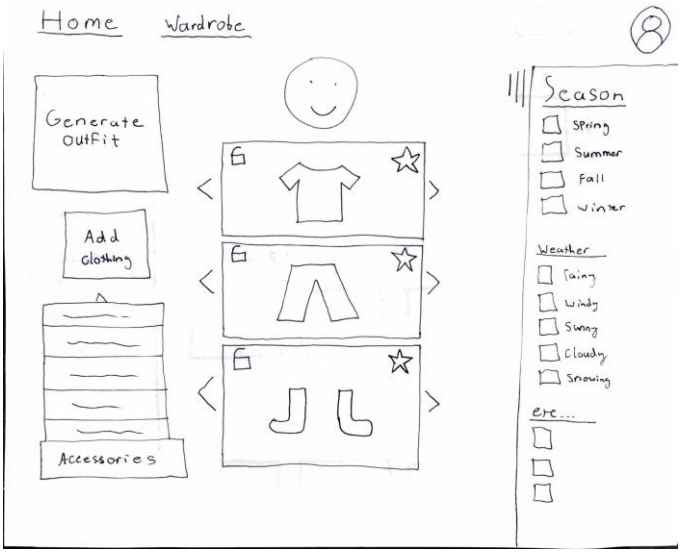



Home
Wardrobe
Season
Style
Fav only
items
Pants >
Shirts >
Jackets>
Shoes >
hoodies/jumpers >
hats >


home
wardrobe
filter:
seasons
clothes
Shirts
Jackets
Pants

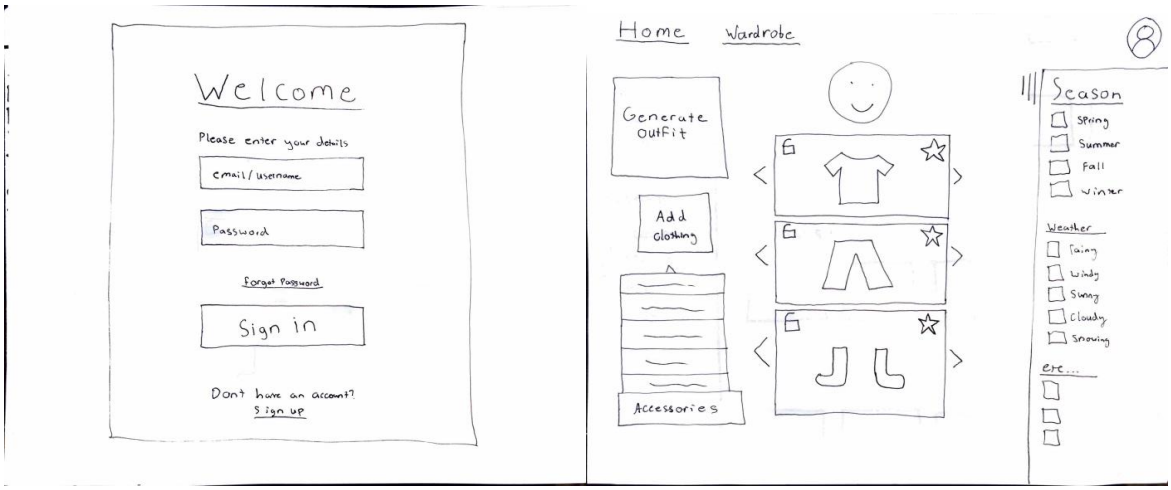

Home
wardrobe
Shirts
Jackets
Filter
season
shirts
jackets
etc

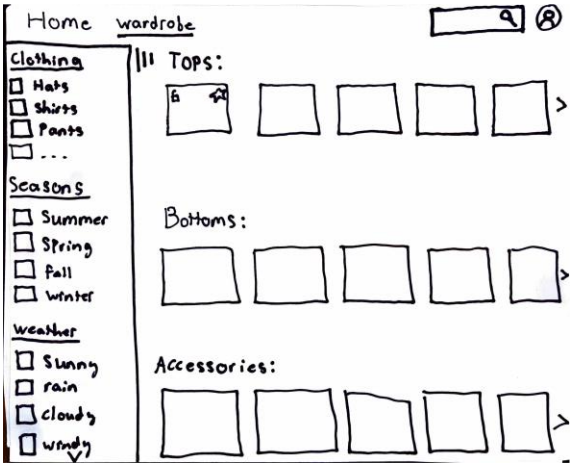
Home page:


Home
Wardrobe
Generate outfit
Add clothing
Accessories
Season
Spring
Summer
Fall
winter
Weather
Rainy
Windy
Sunny
Cloudy
Snowing
etc...

Login page:

Final Designs:





Description of Wireframes/Prototypes

Wardrobe will consist of 3 different pages. These pages are the Login page, the Home page, and the Wardrobe page. The final prototype designs for each page are shown in the Wireframes/Prototypes section under Final Designs.

When you first open the website, you will be placed on the login page. This page consists of two text fields to enter your email or username and password, as well as a forgot password navigation that will allow the user to make a new password. Once all the information has been entered, the user will click sign in and they will be brought to the home page. If they have not visited the site before, they can make an account by pressing the register navigation on the bottom.

On the home page, the user will be able to generate outfits based on clothes that are currently in the system. In the middle of the page are three clothing windows that will display the item that is generated. If the user prefers to cycle through their available clothes to select an outfit instead, there are arrow keys to the left and right of each clothing window that will allow for manual selection. There is a lock icon which will prevent that clothing item from being changed. If the user is looking to create an outfit for a particular clothing type/season/style, they will be able to do so using the filter on the left of the screen, whereby checking boxes they will be able to limit the generation to items with those tags. Lastly, if the user wishes to view all the clothes they have added, they can do so by clicking on the navigation link titled Wardrobe in the top left.

On the wardrobe page, the user will be able to find and look through all the clothes they have added to the site. The clothes will be split into three groups: the tops, the bottoms, and the footwear. If the amount of clothes the user has exceeds the size of the page, they will be able to slide right/scroll to move hidden clothes into view. Similar to the home page, there will be a filter section on the left that will allow the user to hide/show clothes based on what is selected. If the user wants to return to the home page, they can click on the navigation link titled Home in the top left.

Must-Have Features

- Login page, username and password, sign up
- Wardrobe page where you can look at all your clothes separated into correct categories with filters that will decide the content in categories

- Home page where you can look at the outfit generator and switch in and out different parts of it. Similar filtering is available for the generator.
- Clothing type/Season/Style are options for the filtering where we would have checkboxes to be clicked in each.
- Categorize items: Clothing object will have keys of Clothing Type, Season, Style and values will be designated options in the sidebar
- Manually select a piece of clothing by clicking on the lock symbol and generate the rest of the outfit, or generate a full outfit from scratch

Optional Features

- Machine learning / color theory to generate nicer outfits rather than random selection based off categories
- Circular window above clothing windows on home page that will allow the user to upload a picture of their face if they desire to do so.
- Favorite button on home page.
- A way to make sure pictures that are uploaded are clear and fit the correct descriptions
- An add clothes button that is located just below the generate clothes button and will allow the user to take pictures of clothes and add them to their wardrobe.
- Ability to change profile details by clicking on the profile icon in the top right

Project Schedule

Weeks	Dates	Topic	Person	Content
6-8	2/23 - 3/15	Front-End	Alexis	Navbar, filter sidebar
			Nick	Home Page main content
			Reid	Wardrobe main content, log in page
8-11	3/15 - 4/5	Back-End	Alexis	Home App (filters), Wardrobe App (filters)
			Nick	Home App (lock, arrows, images)
			Reid	Login App, Wardrobe App (images)
11-14	4/5 - 5/1	Databases	Alexis	Server Setup, Add and Delete
			Nick	Generate outfits using filters and the user's wardrobe.
			Reid	Authentication

Feedback and Recommendations Addressed:

Project Milestone:

File Uploads: We are making the group decision to not have any file uploading on our website, but instead, we will supply a catalog ready for users to select from.

Filter options: We are making the decision to limit our filter options to only Clothing type/ Season/ style. We chose clothing type as we felt separating clothes into just tops, bottoms and footwear meant in case someone wanted more specific clothes they could use that as a filter option. We felt no need for season and weather so we used the more general option of season. Style and event again are very similar so we felt not many would use it for specific events rather than just the style they would be going for on that day.

Favorite Checkbox: We are removing favoriting from our page as it didn't add anything important to the website, so we moved it to an optional feature. It was originally going to be an option that you would be able to only look at your best clothes and then you could edit the list by clicking the star either to turn it on or off. But we felt if the clothes were put up by someone (even if it is just our selection) on the website then chances are they would like to wear all of them so there would be no favorites.

Optional Profile Pic: We moved profile pic into the optional features section

Registration/sign up: We managed to get it onto one page with the login using a sliding window.

Add clothing separate page: Considering we are now not allowing file uploads and just supplying the wardrobe we will not be needing a separate add clothing page.

Easy to use and mobile friendly: Both moved from must-have features into the project description.

Project Submission:

User specific content: based on the feedback to add more user specific content we went ahead with the idea of adding from a clothing catalog so a different user has different owned and unowned items in their wardrobe. We get into more detail of this in the 'Design Process' section when we discuss our new catalog page.

Remember me button: Removed from login page (more details in 'Design Process')

ARIA accessibility and element deciding: Addressed fully in 'Design Process'

Sidebar filters for the home page: We thought about changing either the filter options or the types of buttons on the sidebar just for the home page to allow the generate button to work more.

However it became quite difficult to ensure any of the filters would be enough to create a whole outfit so we struggled on that aspect trying to come up with filter headings. For the filter buttons, something we thought about was trying to switch it to potentially a radio button which ensures that one of the categories is selected and therefore we could put them on for the types of tops, bottoms and shoes. However this also doesn't work very well for us as we want the user to be able to select a lot of the options at once but the radio button only allows one, for example both jackets and shirts could be picked. Our ideal solution was a type of button that is guaranteed to have minimum one option picked but still allows multiple to be selected. We searched online for something like this and couldn't find anything so we decided to keep the filters the same however add an alert with a clear message saying why the generate button wouldn't be working.

Reduce any duplicate methods: streamlined the displayOutfits into one method. Took the next and previous for top, bottom, and shoes and created one next and previous for all 3. Got a bit stuck trying to combine the three into one next and one previous without using a bunch of 'if' statements so we used Claude to help with the combination and provided the link commented out in the code above the two methods. It used some techniques we weren't too familiar with so we checked with it about the use of 'lookup tables' and 'ternary operators (?).'

ARIA-label referencing: Fixed labelling to be unique addressed in 'Design Process'

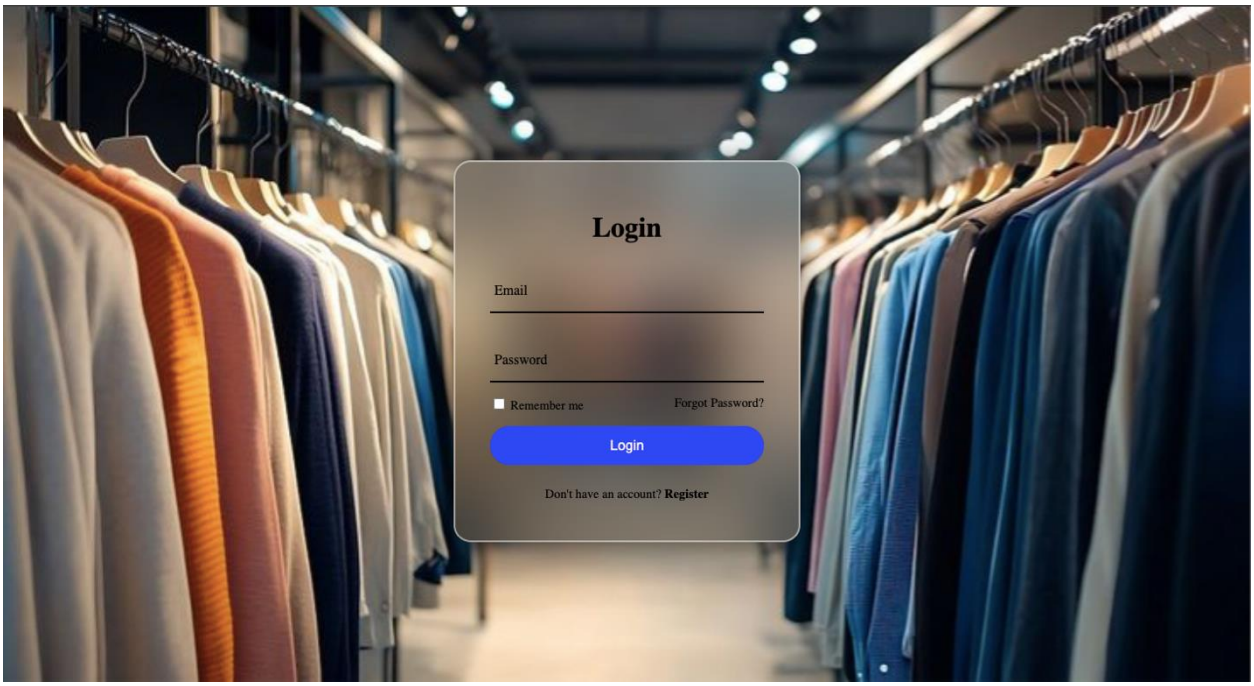
Sidebar semantics: Moved the sidebar into an aside outside of main for each of the html pages

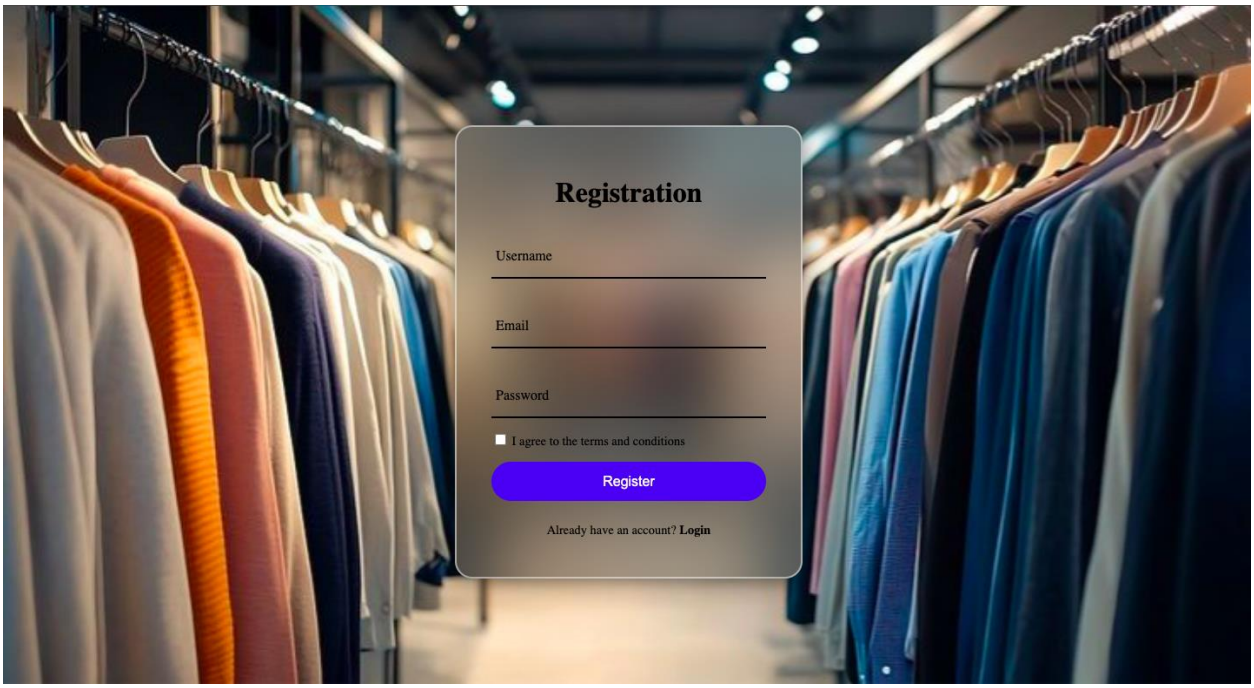
Detail in README: updated the first README describing the structure of our objects in the clothes.json and added all the new files that have been created since the milestone. Added flow and JavaScript functionality in second README.

Comments: added more detailed comments as well as adding comments to stuff we had not learned in class.

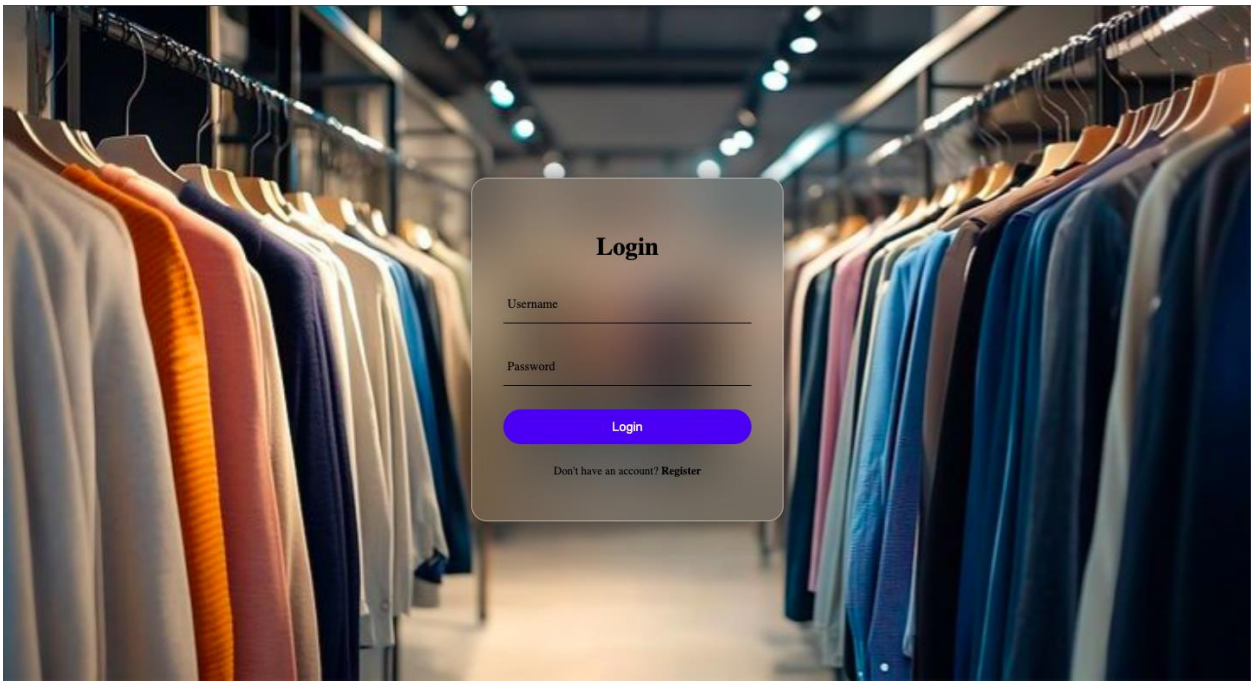
Design Process

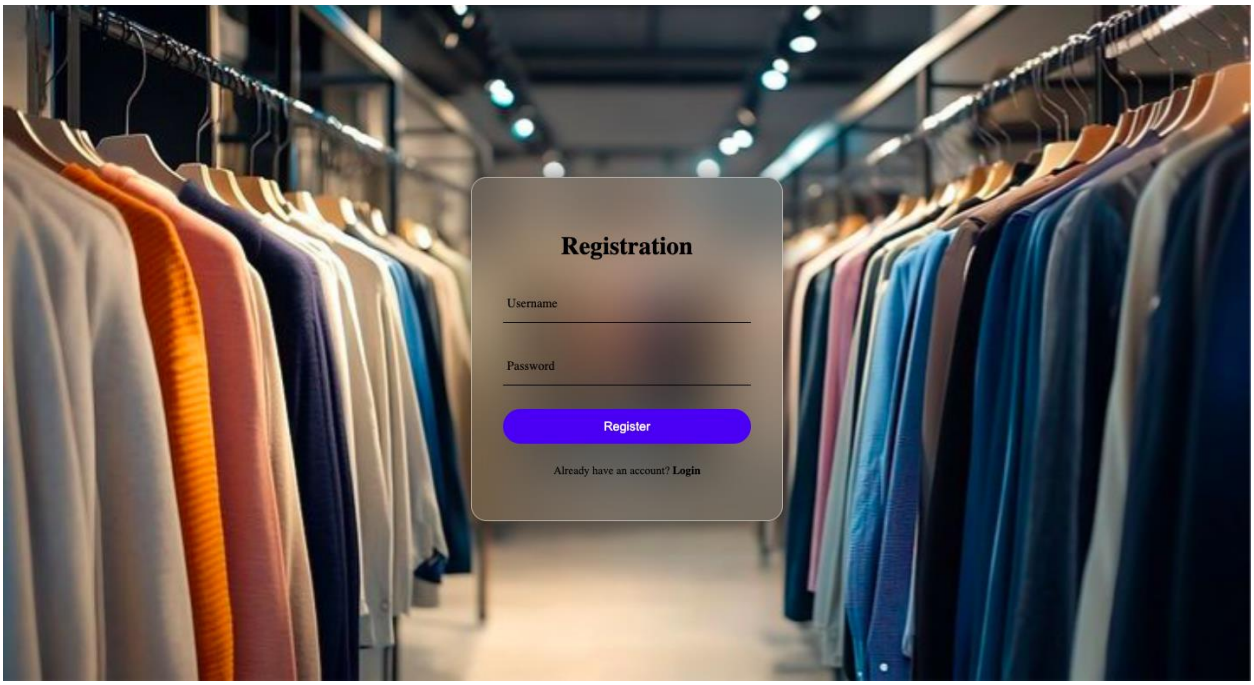
Login/Registration Figure 1:





Login/Registration Figure 2:





About Login/Registration Figure 1:

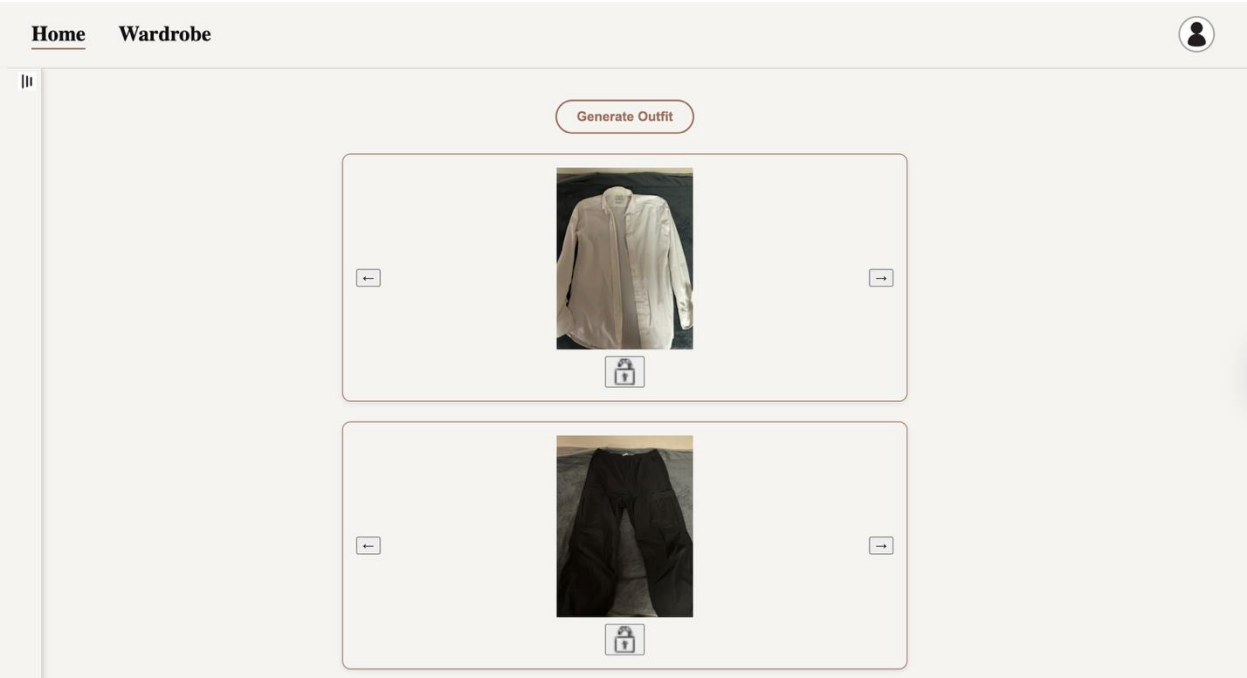
This section of the website is where the login and registration are implemented. On this page, users will be able to sign in to their accounts by typing in their email and password. If they desire to have the system remember them then they can click the remember me box and if they forget their password, they can click the forgot password link to create a new one. If the user does not have an account, they can click the registration link which will replace the login information with

registration. Here they will be able to enter a username, password, and email as well as click on a box that says they agree to the terms and conditions. If the user clicked the registration link by mistake, they could click the login link to return to login. We decided to keep both the registration and the login on the same html to improve the look and flow of the page. This works by having both boxes present on the window with one being invisible and when the button is clicked, the visible box slides off and the other slides in. An issue We had with implementing this was that the registration box was not positioned correctly and as such would not slide into the visible window. This does differ from our wireframes/prototypes. Other than this and a few styling changes the page is the same.

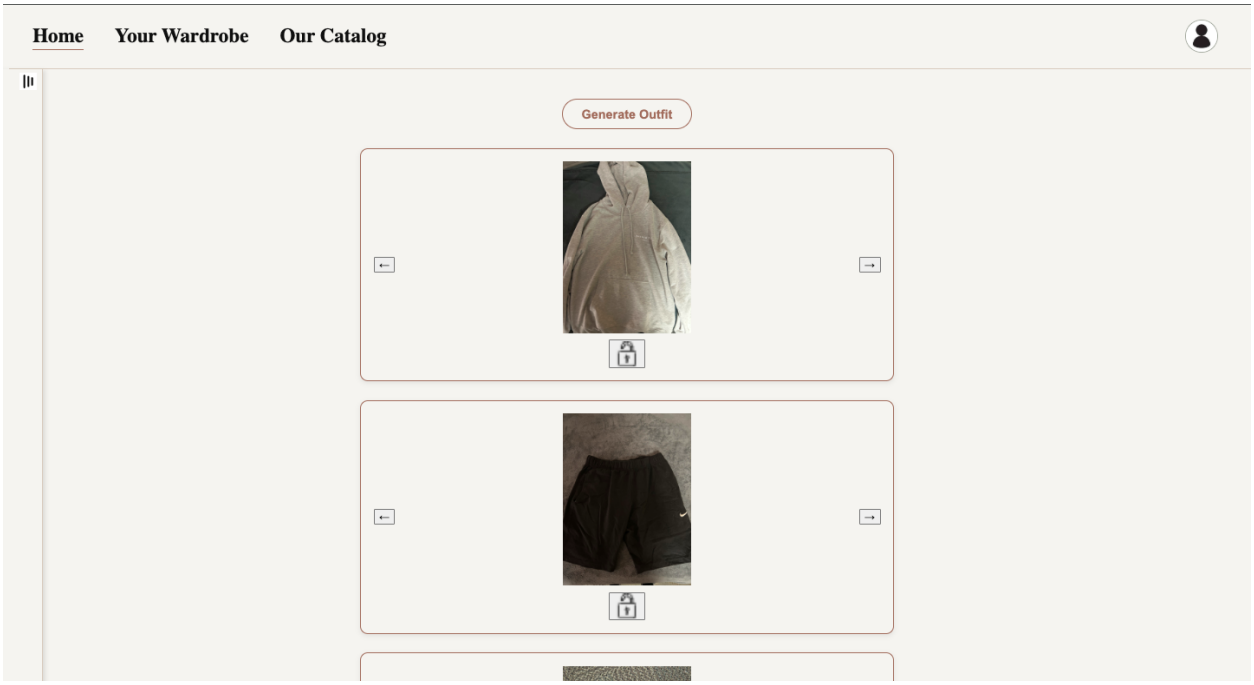
About Login/Registration Figure 2:

Following login/registration figure 1 and the feedback received concerning it, we decided to remove the remember me checkbox, the terms agreement checkbox, and the forgot password link. Our original design included these features as we viewed them as features typically included within pages such as login and registration. However, as we thought about how they were benefiting our project, we decided that they were unimportant to the overall design. We found that they only subverted our attention away from more important aspects of the project. As such, we came to the decision to remove them entirely. Along with this, we decided that the user would not have to enter an email to register and changed the login requirement to be their username and password. Our original design of including the email followed the same thought process as the checkboxes and link. As for why we decided to remove it, we realized that having the user enter their email served no real purpose for our website as we were only using it as a username and as such only made the login more complicated than it needed to be. By removing the email, we were able to use a HashMap in our credentials Json to help with speed of look up as well as make the user registration experience less complicated.

Home Page Figure 1:



Home Page Figure 2:

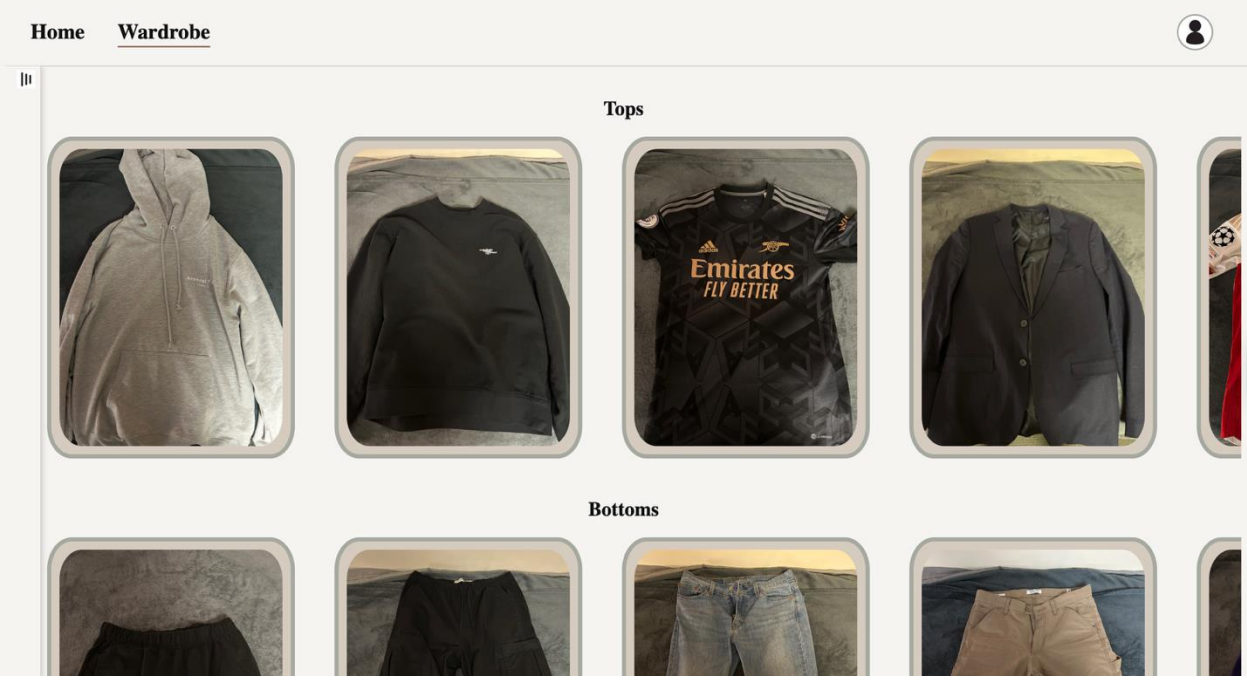


This section of the website is where the Outfit Generator feature is implemented, allowing users to create and customize their own outfits by selecting different items from the given categories of tops, bottoms, and footwear. The outfits get generated by generating a random number and then using that number to select items from an array. We kept getting outfits with items that didn't match, forcing us to click the generator bottom again. So, to ensure that outfits were good, we

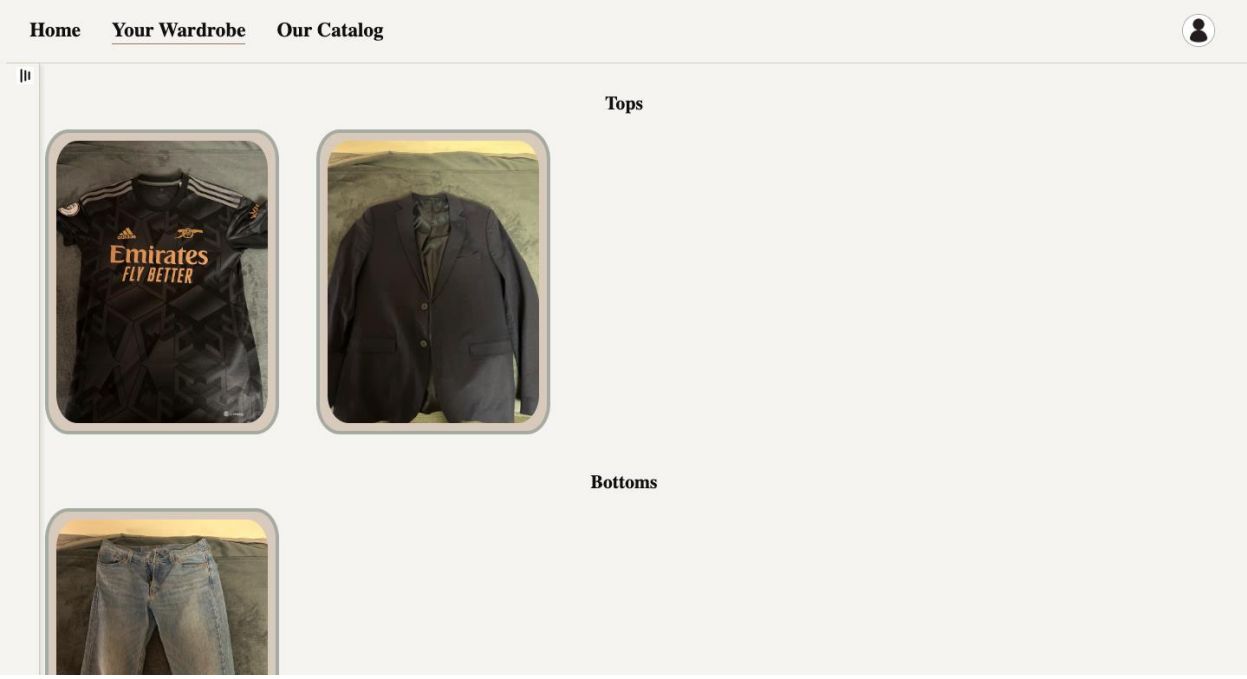
implemented a filter checker. After generating the random number, we check that all the items have a matching filter like style and season. The structure of this section is divided into three different slots, representing the different categories of clothing, with each slot serving as a selection area for the user to choose their desired item of clothing. The user can navigate through different clothing items in each slot by using the left and right arrow keys, as well as display the selected item by using the image element. Each slot also has the option to lock the selected item, allowing the user to keep it the same even if they are navigating through different clothing items, as indicated by the lock icon, as well as the labeled aria-pressed attribute. The accessibility of this section is also considered, as the aria-label attribute is given to all the different buttons to enhance the user experience. This feature is different to our wireframes as we are no longer including an 'add clothing' button, the favorite button has been removed, as well as we moved the generate button to the top due to the sidebar now being on the left side and removing the accessories selector.

The early challenge was choosing the right layout approach for the outfit generator. Since the three clothing slots, top, bottoms, and shoes, were stacked vertically one after another, Flexbox with flex-direction: column was the natural fit, as it excels at arranging items along a single axis. Grid would have been more appropriate if the slots needed to align across both rows and columns simultaneously. For the slot containers themselves, a <section> element with the id outfit-generator was used rather than a plain , because it gives the outfit builder a meaningful landmark that assistive technologies and search engines can identify as a distinct region of the page. Aria-label attributes were also added to the arrow and lock buttons since they contain no visible text. Without them, a screen reader would have nothing meaningful to announce to the user, leaving those controls completely ambiguous. So we understood now that our generic labelling interferes with this process and have gone back into the code and given more unique labelling to each of the aria-labels.

Wardrobe Page Figure 1:



Wardrobe Page Figure 2:



Wardrobe Figure 1 Description:

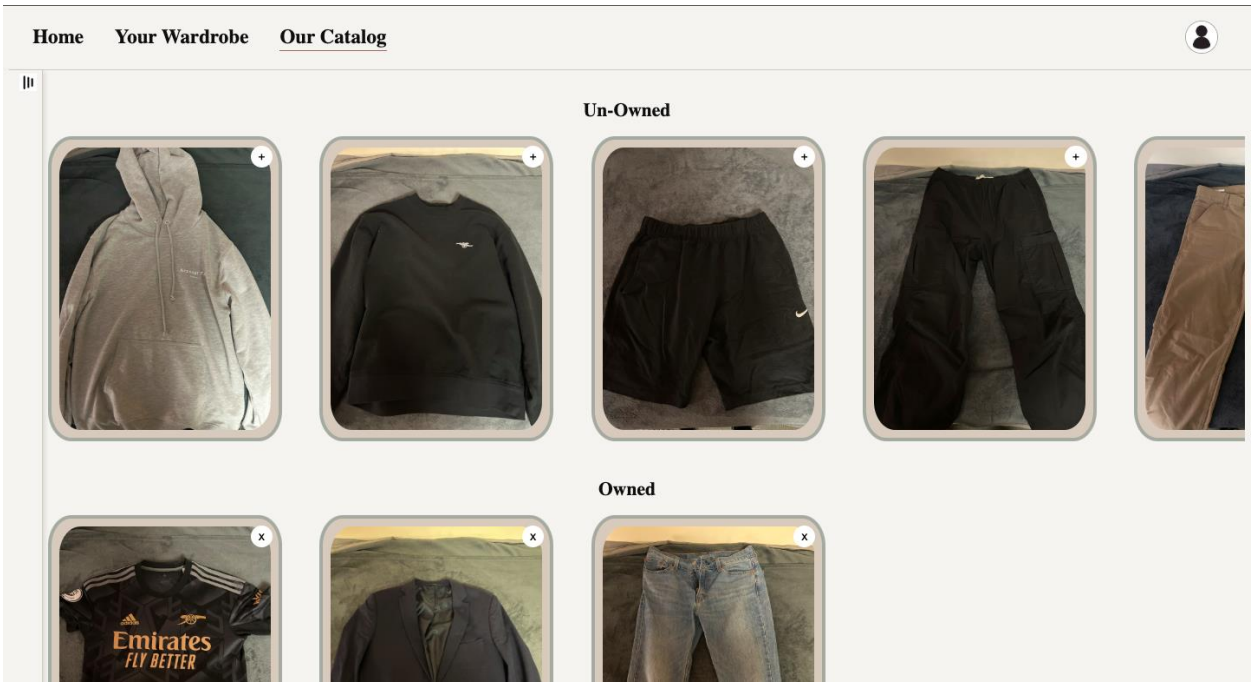
This section of the website is where the wardrobe is implemented. This page allows users to view all available clothes that are currently on their account. The clothes will be split into three sections: tops, bottoms, and shoes. The user will be able to scroll the different rows to see any clothes that are hidden from view. The design of this page is very similar to our wireframe/prototype with a few minor changes to improve usability and to remove unnecessary

features. These changes included removing the favorite, lock, and arrow icons as well as moving the headings to the middle of the page to avoid overlapping with the sidebar. The arrow icon functionality is replaced by the scroll feature to help with showing that there are more clothes than what is visible on the screen. During the implementation of this page, we had some issues with the different rows overlapping.

Wardrobe Figure 2 Description:

The wardrobe page is very similar as to in the project milestone. The only change we made was due to our new catalog page, we allow users to select items to own in their wardrobe. So now the wardrobe will only display owned items instead of all the items. This was a more substantial change as it allows for more user specific content which is the reasoning behind this change. It doesn't solve any old problems but it implements more user specific content which creates a more dynamic website giving more user interaction, hence why this is now better than before.

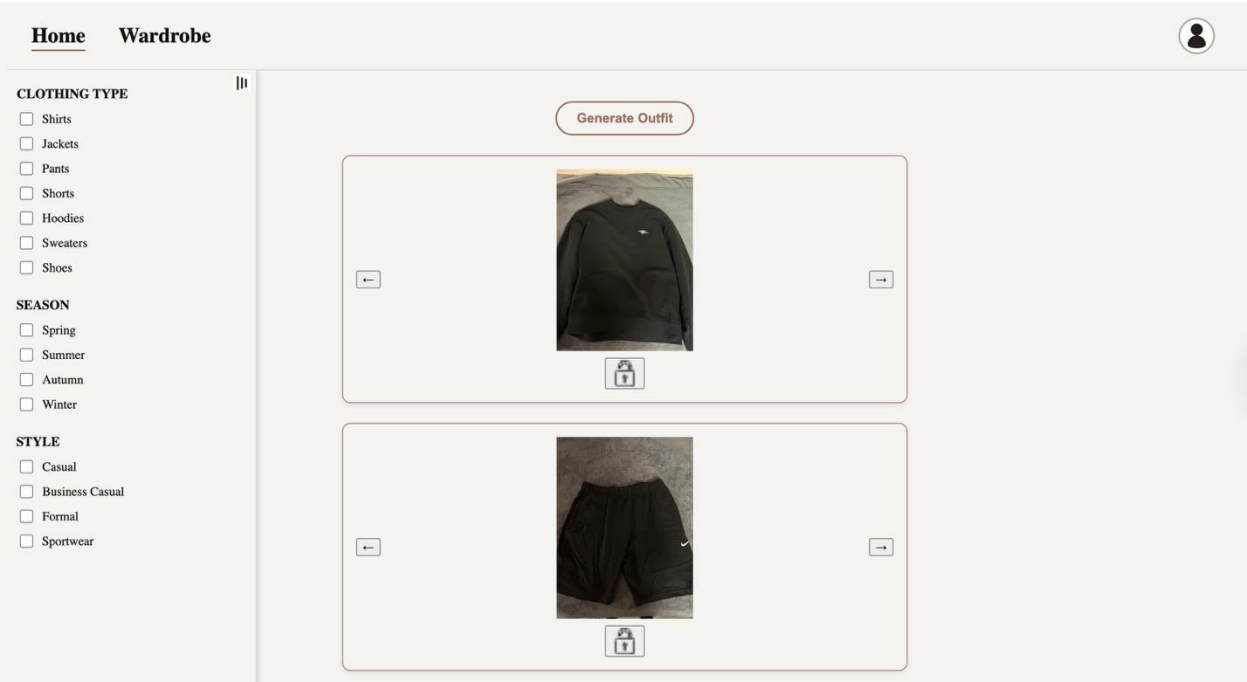
Catalog Page Figure:



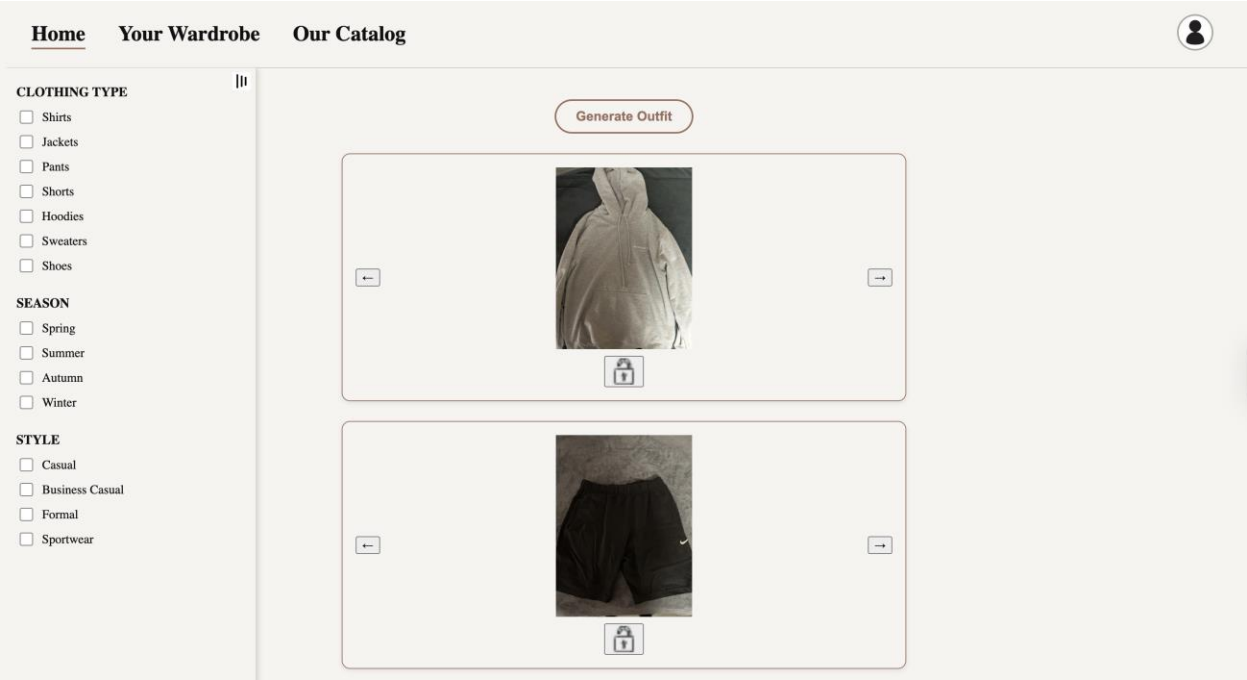
Catalog Page Description:

The catalog page is a new page that we decided to add to our project. On this page, you can see it is very similar to our wardrobe page, however, instead of being split into clothing type it is by owned and unowned items for the current user that is logged in. Additionally we have plus and cross buttons to add and delete from the users owned and unowned items. What this does is it takes the wardrobe content from the specified user in credentials.json and separates the ones that appear into owned and the ones that don't show up in the json into unowned. Once the page has loaded a user can click on the + buttons to move an item from unowned to owned or the x buttons to move an item from owned to unowned. This will update the credentials.json to keep these changes for future signing in as well as for linked ownership onto the other two pages, which now only show the owned items instead of all items. We didn't initially have this in our wireframes as we only decided to add it due to the lack of user-specific content on our website. There weren't many challenges implementing this page of the website as the layout was very similar the original wardrobe page. One thing that came up was I needed to make the items have position relative in order to add the button to the top right of the image which in turn caused the scrolling of the images to be on top of the navbar so I then also increased the z-index of the navbar to make sure it would always be in front of everything. This was something that gave me some confusion at first but I was able to work it out. I struggled a lot with the server.js /save-wardrobe as we had to update our json instead of just adding to it like we had done in class but managed to work out most of it on my own. I had to use a bit of help on Claude with sending the response status as we also hadn't used this in class but without it my page would get stuck pending and would become extremely slow due to it waiting indefinitely for a response that I wasn't sending. I will provide the chat log I used for this it isn't a very big one as I had most of the work done just added the res.status.

Navbar and Sidebar Figure 1:



Navbar and Sidebar Figure 2:



Navbar and Sidebar Figure 1 Description:

We have also implemented the nav and sidebar these are both on the Home and Wardrobe page. Starting with the nav bar, it is always fixed to the top of the screen and allows users to switch between both of our pages, it currently has a small profile section but this doesn't add anything functionally so we are considering removing it. This was a conclusion we also came to with the search bar, which was originally on our wireframes but we thought about it and it wasn't doing anything and would be a waste to properly implement it so we ended up removing it. There wasn't many challenges implementing the nav bar as it is similar on most websites, so I reused some code that I had written before. For the sidebar, it slides in and out of the screen from the left side by clicking on the toggle button at the top right of the sidebar. It allows users to click certain checkboxes to filter which clothes appear on the page based on clothing type, season, and style. These categories are different to our wireframes so we came together as a group and decided on these ones as our final choice as we feel this gives user a good selection factor for all their clothes further described earlier in the address of feedback and recommendations. We also decided to change the sidebar's direction that it comes in. On the wardrobe page wireframe it is shown on the left but on the home page wireframe it is on the right so we decided it made more sense to have them both come in the same direction, as it would not only be much easier to implement but also it helps usability, as it is repeated so users won't get confused. There were some troubles implementing the moving of the sidebar which required some extra research which wasn't too hard to implement once done as well as the positioning of it and the toggle button so it was able to be clicked while still hidden. The filtering works extremely well on the wardrobe page less so on the home page as the home page requires you to have at least one option for each slot to work so the filters chosen have to accommodate for that. This is something that will work better when we add more clothes to our JSON.

Navbar and Sidebar Figure 2 Description:

Navbar: For the navbar we had to add an extra <li> element to it in order to accommodate for the adding of 'Our Catalog' page as well as then we had to change the wardrobe page to 'Your Wardrobe' we did this to show the user that the wardrobe was customizable so when they originally go onto a blank page after registering they will hopefully realize they have to add items from 'Our Catalog'. Additionally we addressed our concerns about the profile pic and decided to keep it but as discussed with you in class we have made it so when clicked the user will return to the login and registration page. We changed the navbar elements to allow for our update of more user specific content so it was a simple but substantial change as without that change there would be no access to our newly created catalog page. It doesn't solve any real problems, but it helps with the issues we had without our user-specific content. With the profile-pic it was a very simple change just wrapping the image in an <a> tag and having a href back to the index.html so nothing cosmetic but a very easy substantial change just making the click do something, in this case taking you back to the login/registration page.